NeuroSonic

Offline spoken walking-path awareness and navigation,
iPhone client + laptop GPU

Research prototype · results measured on the laptop, 2026-11-03

> Not a safety device. It does not replace a white cane, a guide dog or human assistance.

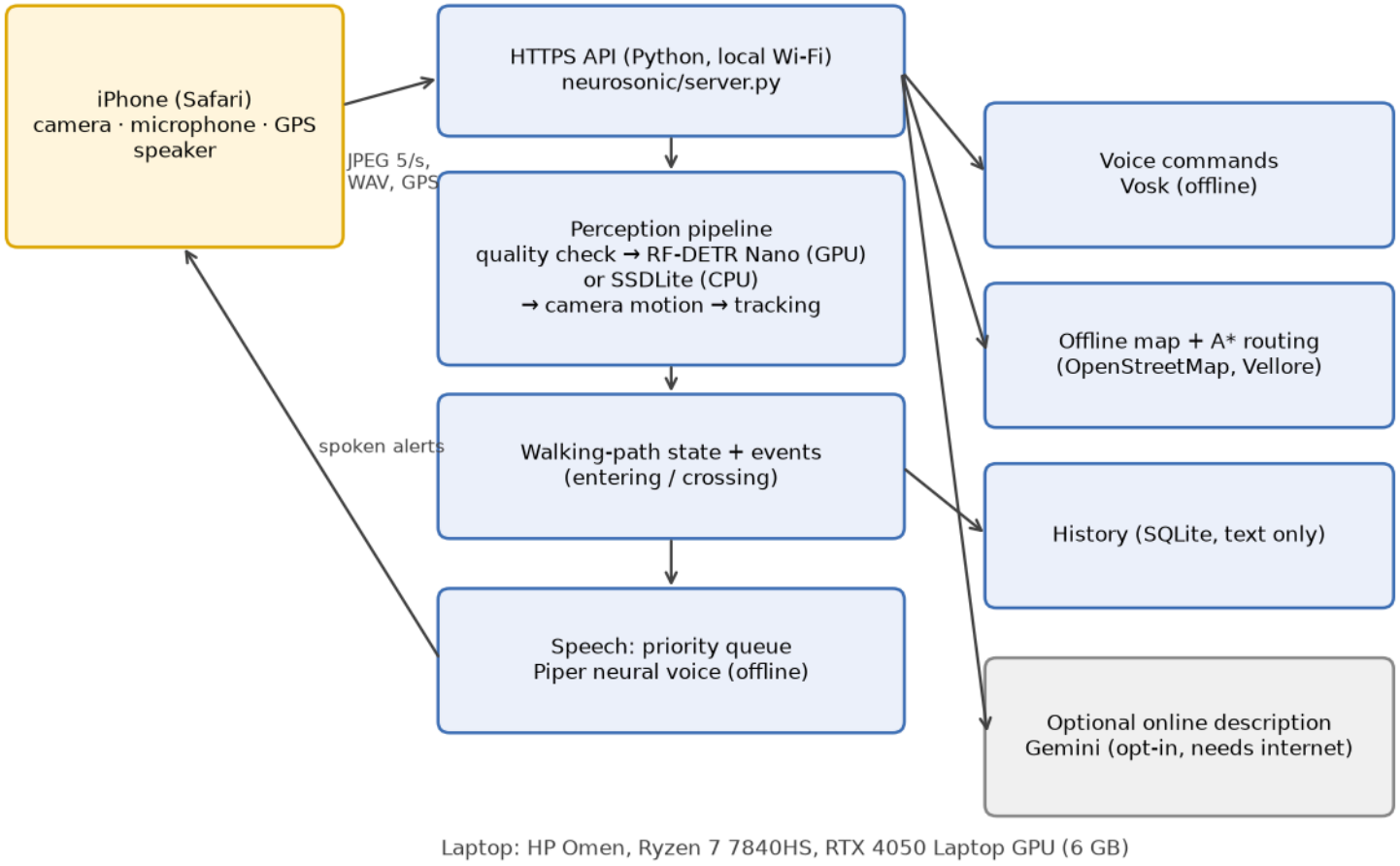
Problem and objectives

Need: a short spoken message when something is in, or moving into, the walking path, plus turn-by-turn guidance, without depending on the internet.

Starting point: WOAD (near-IR + depth model) needs a CS30 sensor that a phone does not have.

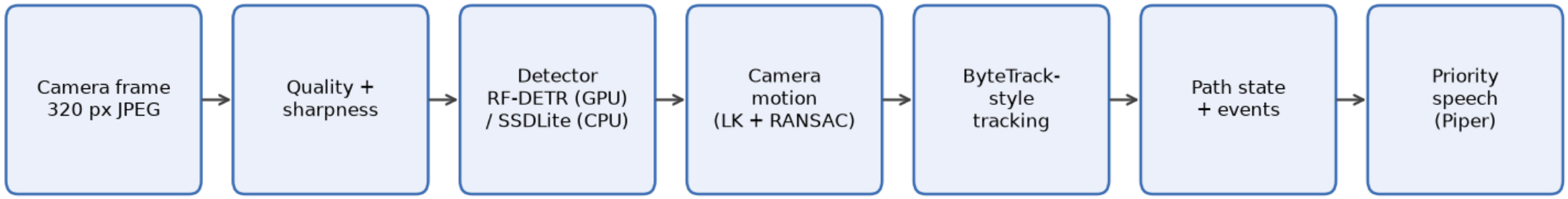
- ≈ 5 frames/s, answer well under 200 ms
- detector chosen by measurement and licence
- movement alerts without false alarms from the user's own walking
- local functions work offline; cloud is opt-in
- no distances; never "safe"

Architecture



Phone = camera, mic, GPS, speaker (Safari). Laptop = all AI, over local Wi-Fi.

Per-frame perception pipeline

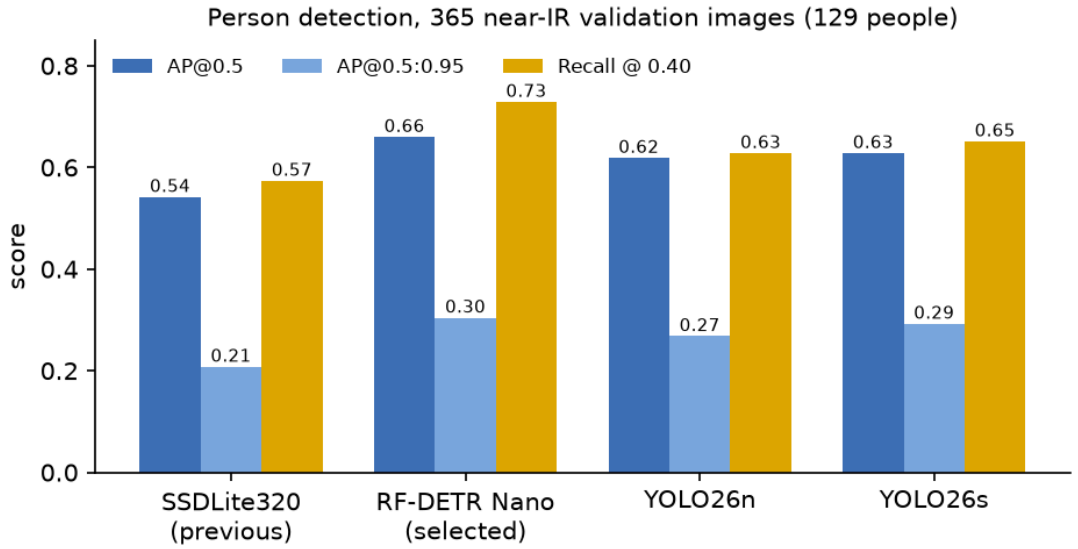


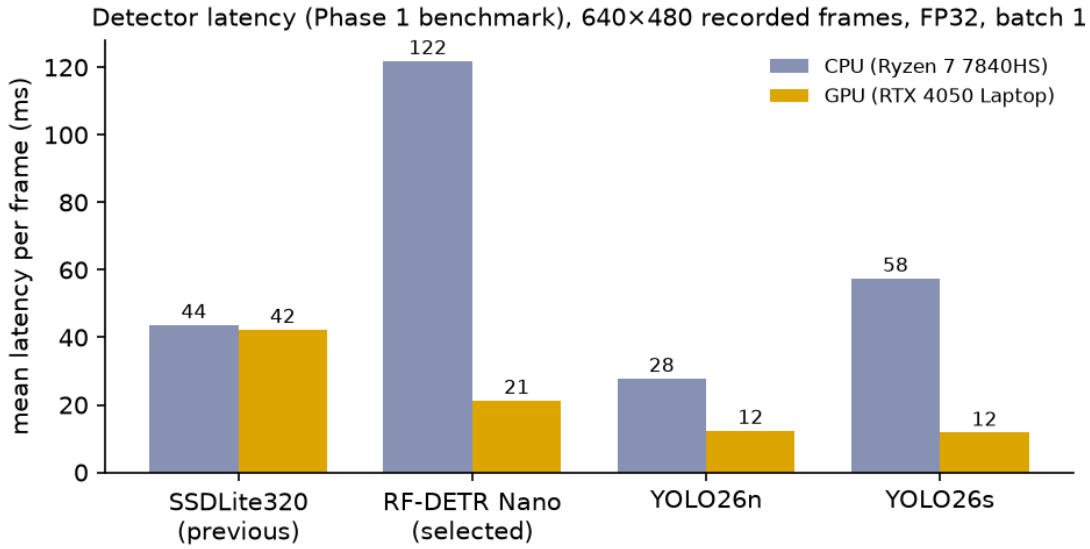
Laptop, 5 frames/s, RF-DETR FP16 on the GPU: detector median 53 ms (GPU in its low-power state between frames);

request median 70–78 ms in the 10-minute soak test; capture → answer median 67 ms in the browser test (loopback, no Wi-Fi).

- RF-DETR Nano on the GPU (FP16), automatic fall-back to SSDLite on the CPU
- camera motion from optical flow + RANSAC → tracks move *relative to the background*
- walking-path states confirmed over 2–3 frames; spoken only on change

Detector selection: same images, same harness





RF-DETR Nano: best person AP@0.5 (0.660) and recall (0.729), 21 ms on the GPU, Apache-2.0. YOLO26 is faster but AGPL-3.0 and lower recall. Data: 365 near-IR images, 129 people (not phone images).

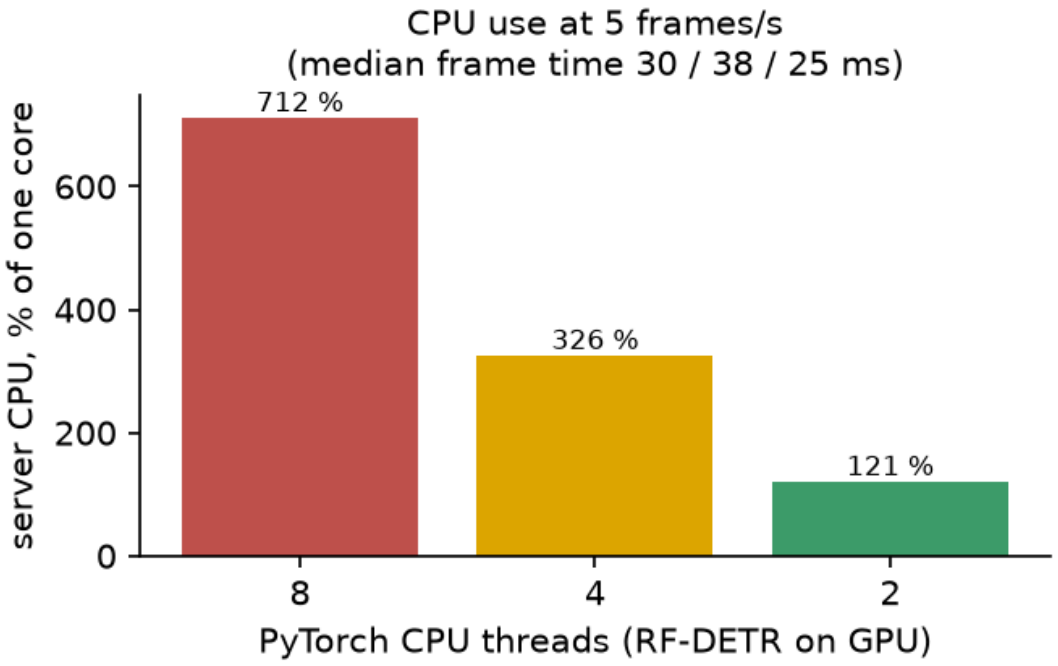
outputs/benchmarks/phase1/results.json

Movement events: conservative by design

- **Entering:** outside the path 1.5 s ago, inside now, moving towards it ≥ 0.12 widths/s
- **Crossing:** centre < 35 % → > 65 % (or back) within 3 s
- only people, vehicles, animals; confirmed tracks (≥ 3 updates, ≥ 0.4 s)
- silent while the camera shakes or zooms, once per track, ≥ 3 s apart

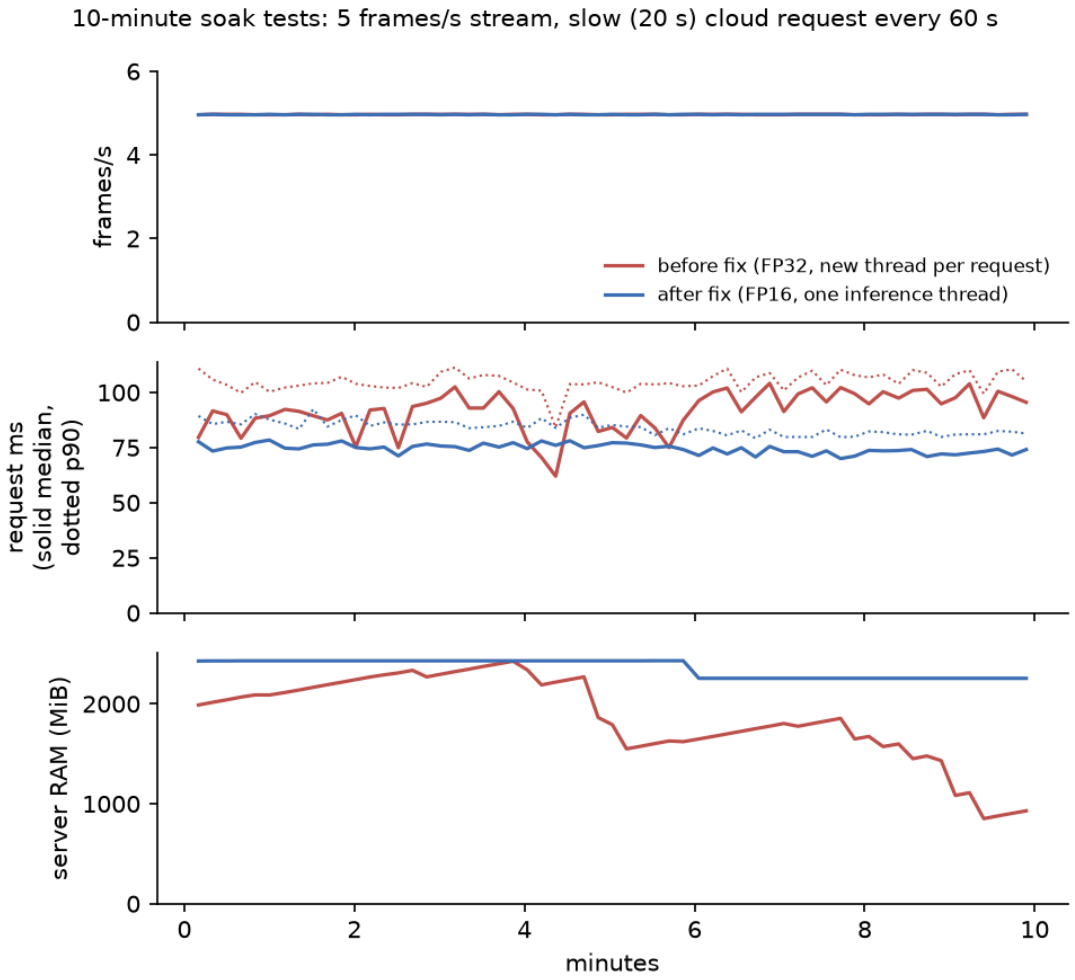
- "Caution. A person is moving into your path from the left."
- "A bicycle is crossing in front of you, from left to right."
- No "approaching": box growth is mostly the user's own walking.

Serving at 5 frames/s: three measured problems



- **Idle CPU threads:** 712 % → 121 % of one core with 2 PyTorch threads
- **GPU power state:** at 5 fps the GPU drops to P8 (~390 MHz): same inference ~16 ms → ~74 ms
- **FP16:** at 5 fps median 66.5 → 53.2 ms, same person AP@0.5 (0.660)
- **One inference thread** instead of a new HTTP thread per frame

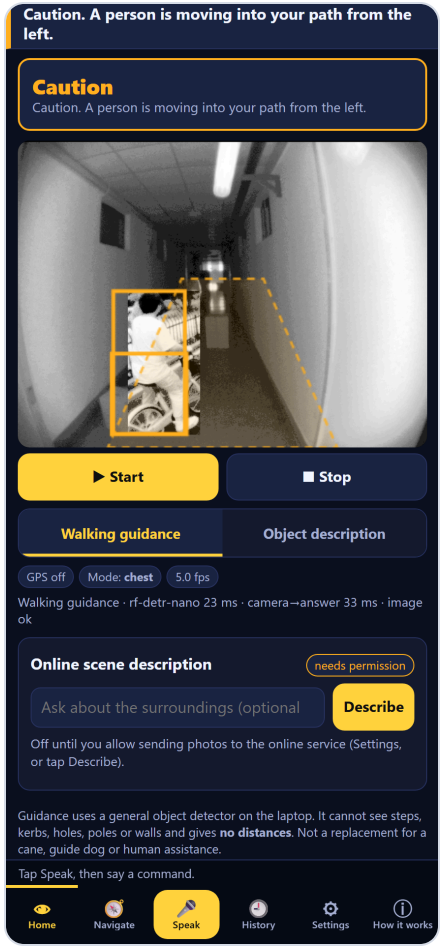
10-minute soak tests



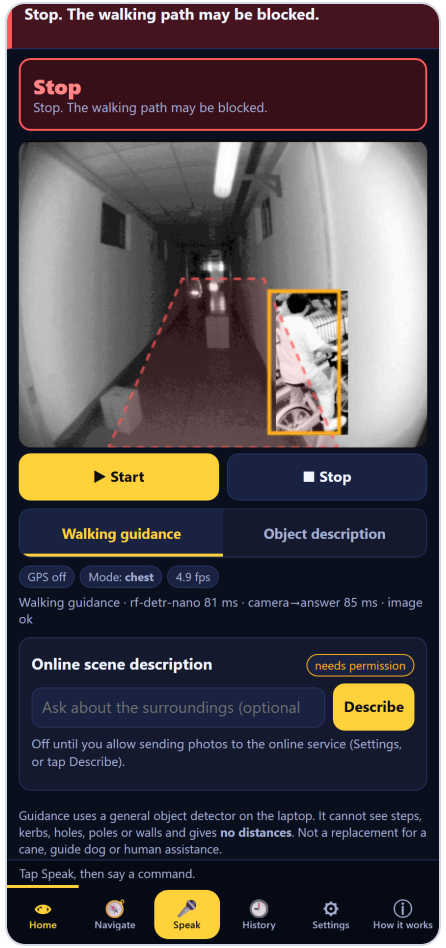
- 2980 frames each, **0 errors**, 4.97 frames/s
- request median 70–78 ms after vs 62–104 ms before
- worst p90 92 ms (was 111 ms)
- no memory growth trend; GPU memory flat
- 7 / 7 slow (20 s) cloud requests answered while frames kept flowing
- laptop loopback, not Wi-Fi; 10 min cannot rule out a slow leak over hours

tools/soak_test.py · outputs/benchmarks/soak/

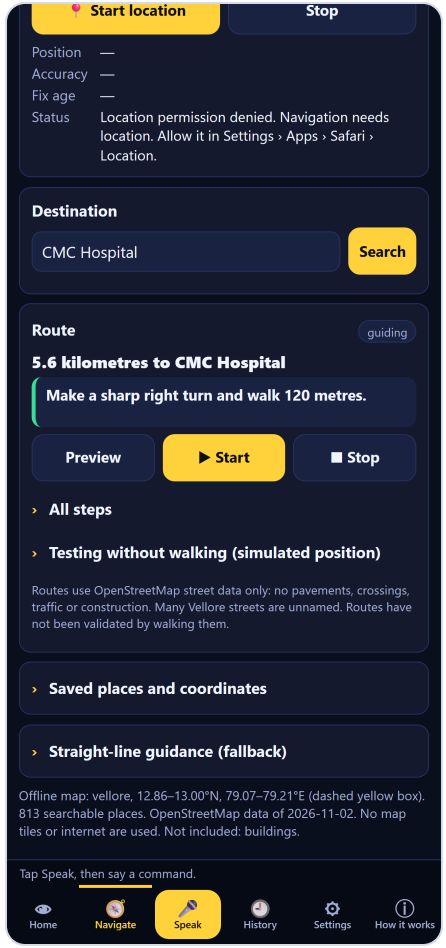
The phone interface (Chrome, iPhone-size, fake camera)



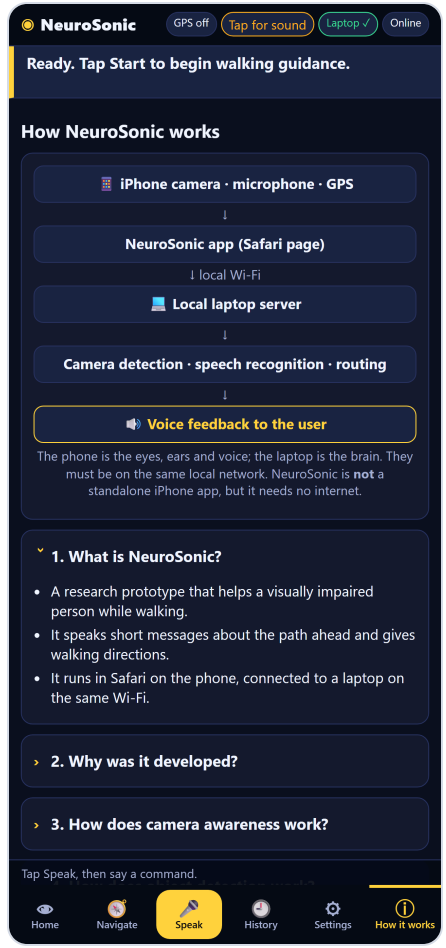
entering event



"Stop. The walking path may be blocked."



offline route, spoken turns



How it works

Camera images: recorded cyclist composited over a recorded corridor, not field footage

Testing

204

Python + Node tests: OK in both environments (GPU and CPU-only)

7 / 7

headless-Chrome tests at iPhone size, network blocked

67 ms

median capture → answer, 4.91 frames/s (loopback)

- event → neural voice starts in 257 ms; "still waiting" spoken at 15 s during a slow cloud reply
- no event for a static car while the camera pans, a shaky camera, slow drift, or chairs
- quality gate: GPU and CPU launches, frames, speech, search, routing, voice command

Limitations: what is *not* shown yet

- **No iPhone or field validation:** RF-DETR on real iPhone frames, Wi-Fi latency, streets, users
- Detector quality from near-IR proxy data; live tests use a fake camera and a composited clip
- 23 COCO classes only: no steps, kerbs, holes, poles; no distance; no "approaching"
- Phone needs the laptop and Wi-Fi; Safari stops the camera when the screen locks
- Cloud description: the one approved live request timed out

> Supplementary information only. Use with a cane and, in tests, a sighted companion.

Next steps

1. Evidence
(docs/IPHONE_VALIDATION_PROTOCOL.md)

- 23 scenarios × 5 trials: path states, events, low light, blur, navigation, GPS, voice, cloud
- record results in docs/validation_results_template.csv

2. Then capability

- monocular depth → "approaching" alerts
- kerbs, steps, potholes with openly licensed data
- native app with on-device inference